

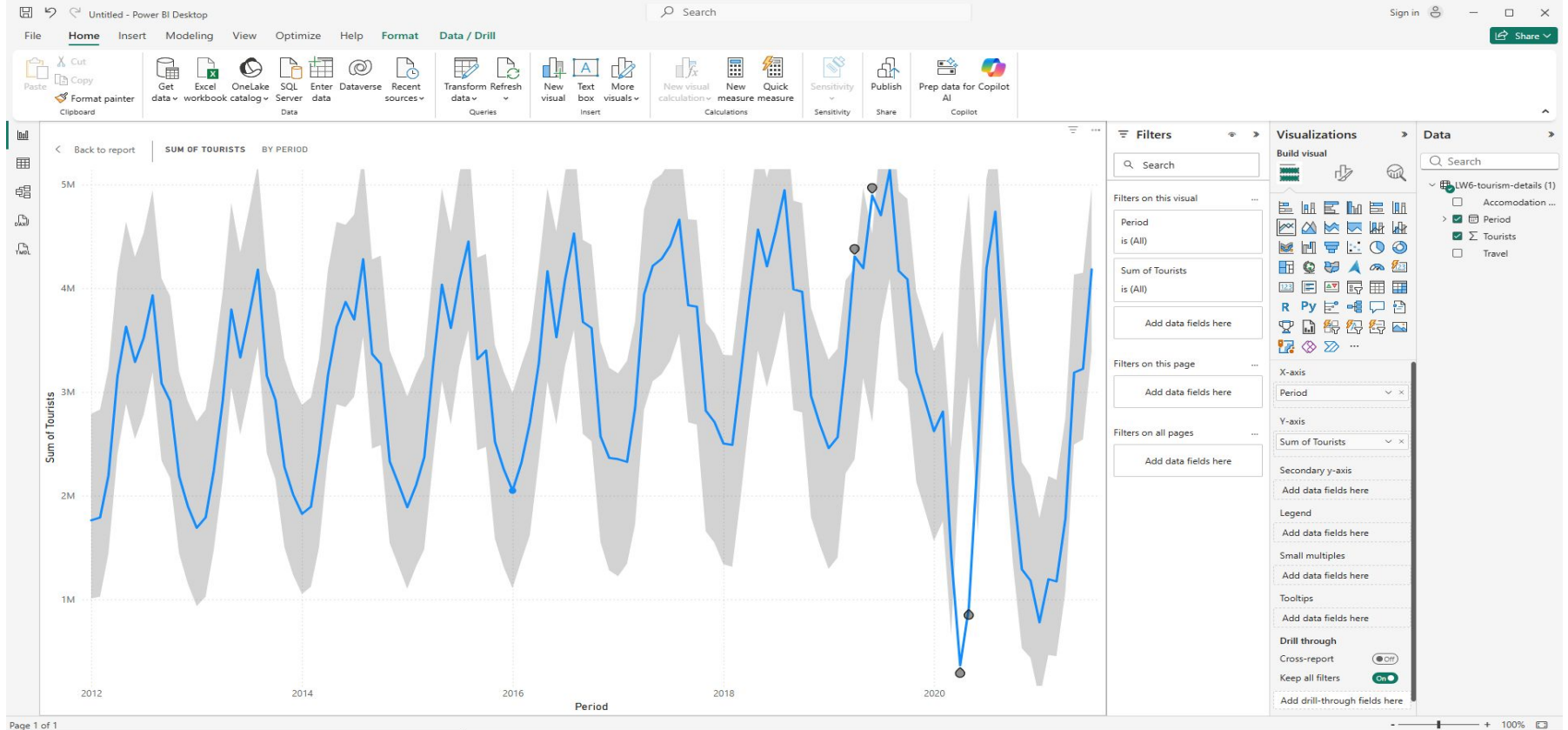
Laboratory Work 6 Activity | Detecting Anomalies in Time-Series Data Using Power BI

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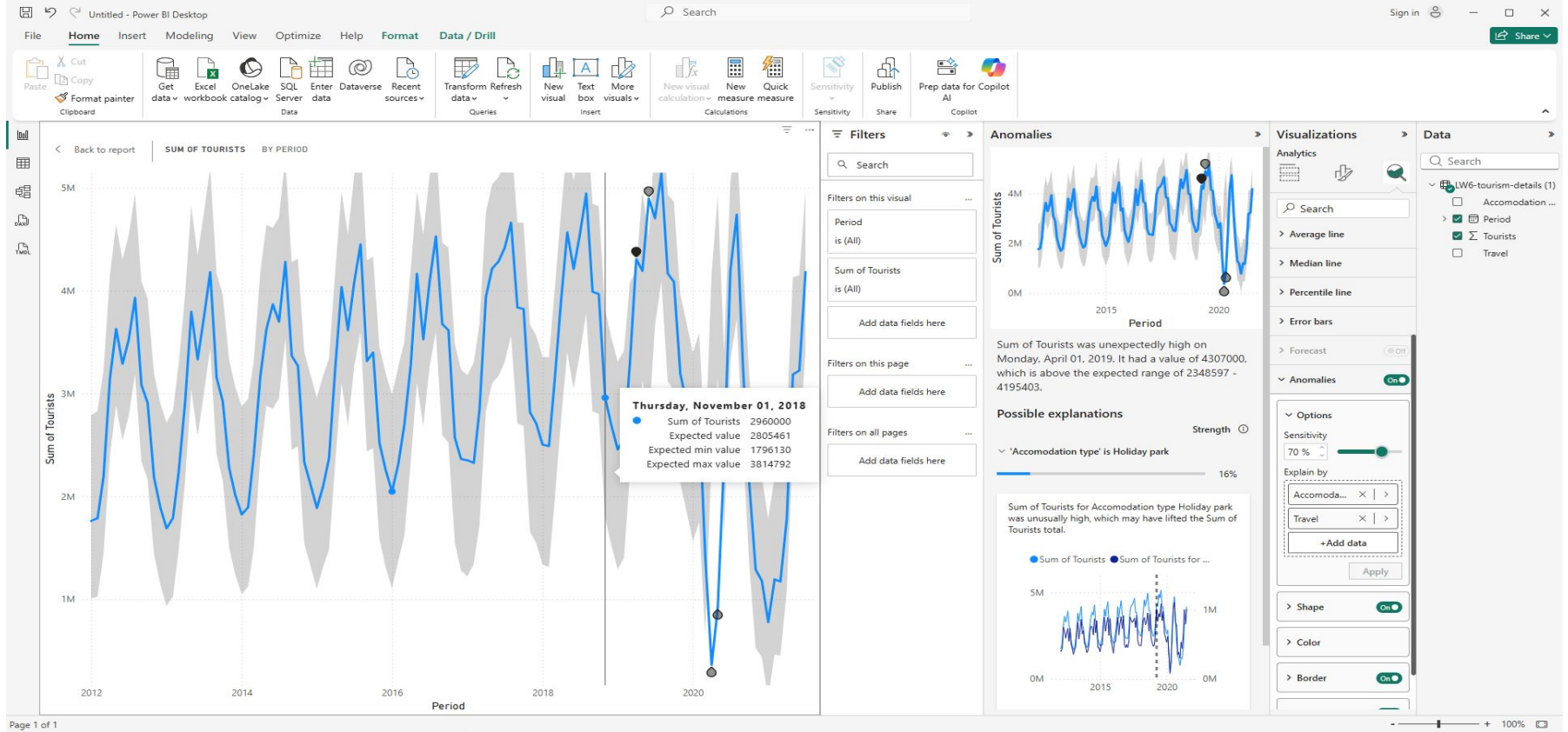
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IS 107 - Business Intelligence

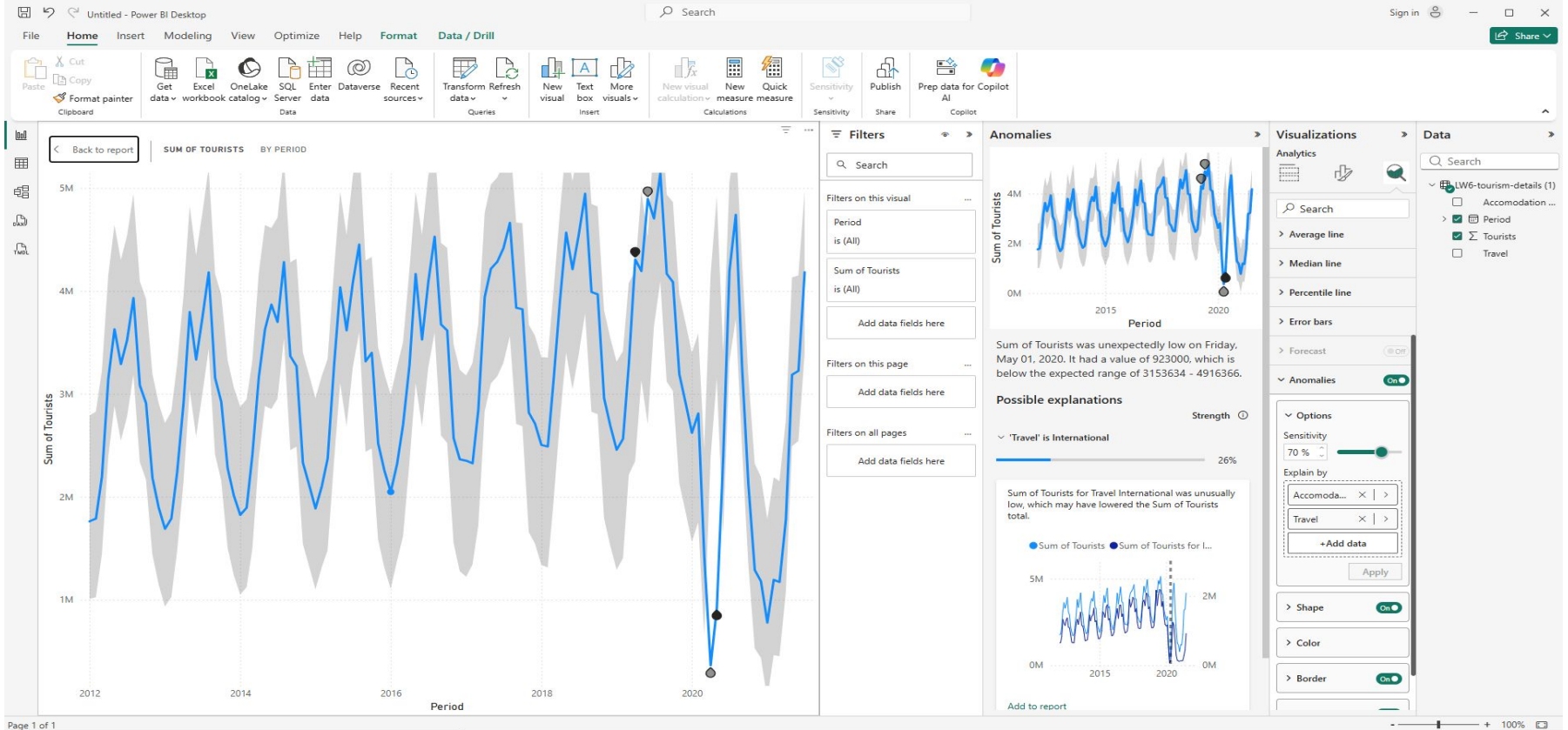
Data Visualization



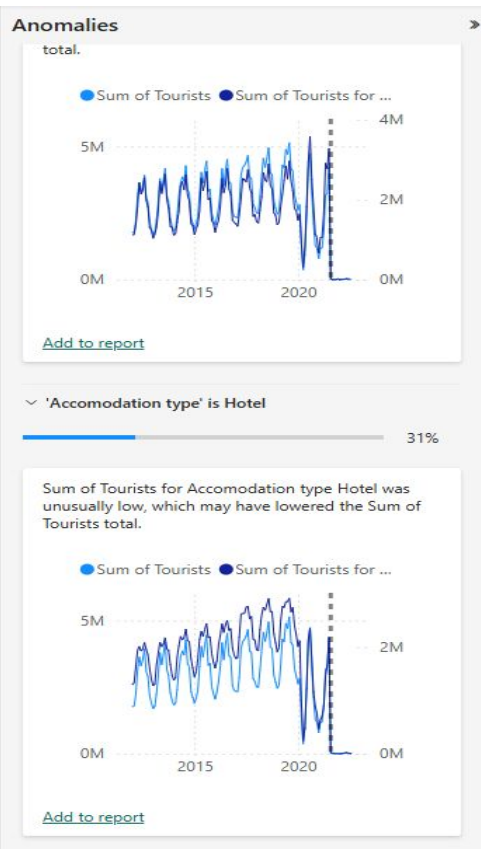
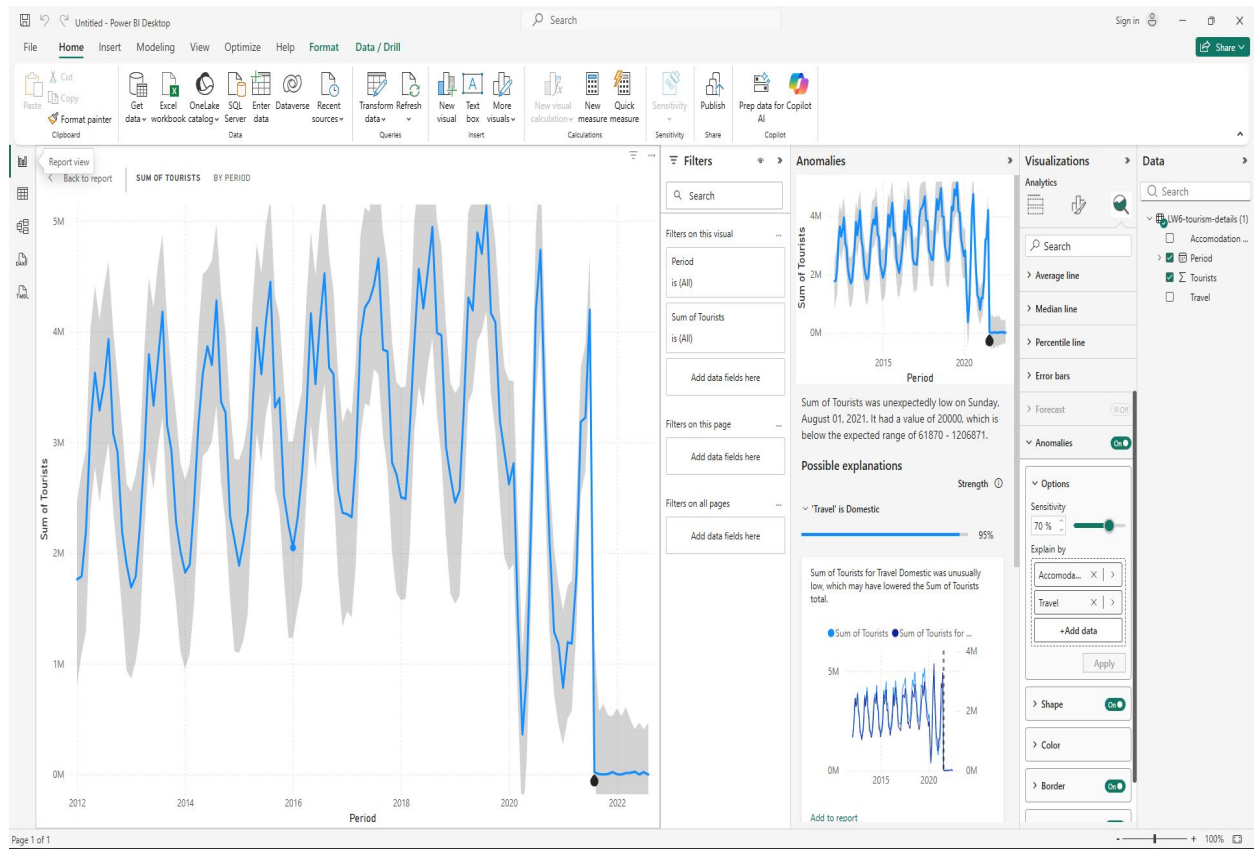
Anomaly occurring in April 2019.



Anomaly in May 2020



Enhancement Task



Interpretation and Analysis

What trend and seasonality patterns are observed?

Answer: The time-series chart shows a clear and consistent seasonal pattern repeating every year from 2012 to 2021. Tourist numbers peak during mid-year (around July–August) and dip during the start and end of each year. There is also a general upward trend from 2012 through 2019, meaning tourism was steadily growing year over year. This growth then sharply reverses in 2020, bringing numbers down to historically low levels.

Interpretation and Analysis

Which anomalies are obvious vs. subtle?

Answer: The obvious anomaly is the dramatic crash in May 2020, where the sum of tourists dropped to 923,000 — far below the expected range of 3,153,634–4,916,366. This is visually striking on the chart as a sudden deep plunge that breaks the established seasonal pattern entirely.

The subtle anomaly is the spike in April 2019, where tourists reached 4,307,000, slightly above the expected range of 2,348,597–4,195,403. This one is harder to spot visually because it still falls within the general upward trend, but Power BI's anomaly detection flagged it as statistically unexpected.

Interpretation and Analysis

What factors contributed to the anomalies?

Answer: For the April 2019 spike, Power BI identified that Holiday park accommodation was unusually high (16% strength), suggesting a surge in domestic or international visitors choosing holiday parks that month, possibly due to a public holiday, school break, or a seasonal event unique to that period.

For the May 2020 crash, Power BI pointed to International travel being unusually low (26% strength). This aligns directly with the COVID-19 pandemic New Zealand closed its borders in March 2020, which effectively eliminated international tourists and severely suppressed domestic travel as well due to lockdowns.

For the Enhancement Task anomaly in August 2021, the explanation pointed to Domestic travel being unusually low (95% strength), with Hotel accommodation specifically identified as the contributing factor. This is consistent with New Zealand's Delta variant lockdown in August 2021, which restricted movement and caused hotel bookings to collapse.

Interpretation and Analysis

Are the explanations logically valid? Why or why not?

Answer: Yes, the explanations are logically valid:

- The May 2020 explanation citing low international travel is strongly supported by real-world events — New Zealand's border closure was one of the strictest in the world during COVID-19, and the timing matches perfectly.
- The April 2019 explanation about Holiday park demand is reasonable, though less definitive since a 16% strength is relatively weak — it explains part of the anomaly but likely other combined factors (such as overall tourism growth and favorable conditions) also contributed.
- The August 2021 explanation about low domestic Hotel demand is highly credible given the Delta lockdown, and the 95% strength score confirms it as the dominant driver, making it the most reliable explanation of the three.